

FROM KV PROGRESS TO SERVING GOODPUT: MECHANISM AND ADMISSION IN CROSS-VENDOR PD

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ABSTRACT

Faster KV-cache progress need not improve serving when prefills share execution with ongoing decoding. We study this interaction in an 8B cross-vendor deployment pairing Iluvatar BI-V150/vLLM with MetaX C500/LMDeploy. With routes and complete-output replay fixed, we cross native write permission with prefix admission while matching sender ownership and control-read handling. Across two opposite-order blocks, allowing writes during native execution with immediate admission adds 15–17 ms to short-request mean completion TPOT and reduces short-request success under joint service-level objectives (SLOs) from 20/20 to 0/20 per block. Empty-set admission restores these successes but increases long-request mean first-token latency from about 17 to 56 s. With a separately prespecified two-second admission rule, allowed progress retains measurement-cohort write overlap and lowers long-request mean first-token latency by about 31% relative to matched exclusion, while losing one or two short successes; no long request meets the joint target. The progress–admission interaction recurs across three blocks of one new microburst arrival pattern. The progress penalty also persists after removing an extra replay token roundtrip. In a separate native-budget experiment, 128-position caps shorten mean mixed-step host residence but increase cumulative per-request exposure and recover no short successes. These controls establish an admission-dependent runtime effect without uniquely identifying its CPU, GPU or batching mediators. They motivate evaluating KV-progress and prefill-granularity optimizations against complete request-level service objectives.

1 INTRODUCTION

Prefill–decode (PD) disaggregation separates prompt processing from autoregressive generation and transfers the intervening KV state (Zhong et al., 2024; Patel et al., 2024). Across vendors, it connects native runtimes with different execution and scheduling behavior. The serving objective is to turn faster state movement into more requests meeting their latency targets.

When a device serves whole requests alongside remote prefixes, earlier KV progress can change their coexistence. SLO-aware systems recognize competing latency objectives (Wang et al., 2025; Arun et al., 2026); isolating a progress optimization additionally requires controlling placement and application work.

We ask: *how does native KV progress change joint-SLO goodput under fixed routes, and can prefix admission change that effect?* Complete-output replay and frozen routes match work and placement, including warmup. Native witnesses verify payload writes during source-engine work; client events identify gained and lost successes. Figure 1 locates the controlled boundaries. A whole-request control uses the same devices. Matched-ownership and fixed-threshold interventions test admission’s role.

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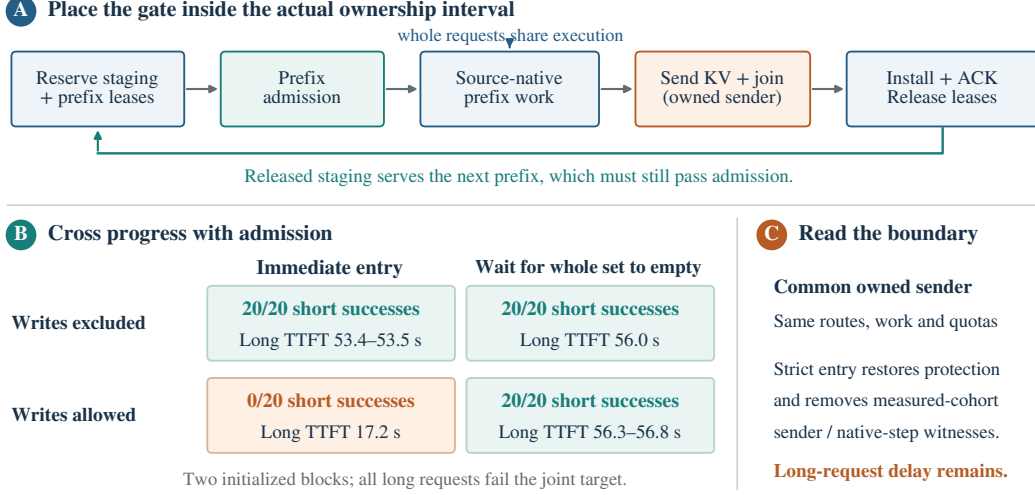


Figure 1: **KV progress connects to later prefix entry through resource reuse.** (A) Staging and prefix leases are held before admission. Completing transfer, installation, acknowledgement and sender join releases the resources for a subsequent prefix, which must still pass its admission gate. The return arrow denotes this implemented dependency, not a uniquely established performance mediator. Whole requests bypass the leases and share source-native execution. (B) A common owned sender crosses write permission with prefix entry. Cells report rounded block-mean long TTFT and per-block short successes. (C) Protection carries a long-request cost.

The paper connects the phenomenon to an executed mechanism intervention:

- **A replicated fixed-placement finding.** Four initialized blocks on two physical allocations show that large long-request TTFT gains can coexist with complete loss of prior short-request successes. CPU-quota comparisons establish persistence at both tested quotas.
- **A matched progress–admission interaction.** A native factorial intervention reproduces the loss with sender ownership held constant. Prefix-entry gating attenuates the progress-associated short-TPOT penalty in both execution orders.
- **Limits of local optimization metrics.** Strict and fixed-threshold entry expose a protection–latency tradeoff. Separately, 128-position caps shorten mixed steps but increase cumulative exposure without restoring qualification: local granularity is insufficient to predict client protection.

The supplement provides the native implementation, workloads and reconstruction scripts.

2 CONTROLLED EXPERIMENT

2.1 CLIENT QUALIFICATION

For request i , let a_i be planned arrival, f_i its first nonempty client emission, d_i completion and N_i the number of visible tokens. One client monotonic clock gives

$$\text{TTFT}_i = f_i - a_i, \quad \overline{\text{TPOT}}_i = \frac{d_i - f_i}{N_i - 1}, \quad T_i = d_i - a_i.$$

For target vector $\tau = (\tau_F, \tau_P, \tau_C)$, qualification is

$$q_i = \mathbf{1}\{\text{TTFT}_i \leq \tau_F, \overline{\text{TPOT}}_i \leq \tau_P, T_i \leq \tau_C\}.$$

Average completion TPOT measures the complete decode interval; a maximum token-gap objective would define a different success set.

2.2 FIX PLACEMENT AND APPLICATION WORK

P0 and P1 share ordered arrivals, the complete warmup/measurement route vector, model, precision and quotas. Each executed destination is checked against the plan. After each native sampling operation, both arms commit the same reference token as subsequent history. This matches complete application trajectories while retaining native queries, sampling and batch-composition effects. Native scheduling overhead remains a measured response. Replay controls cost and timing; natural-answer quality is outside its estimand. Appendix A retains the complete work accounting and Algorithm 2.

2.3 INTERVENE ON SENDER PROGRESS

P0 writes an existing KV payload on the event loop that invokes synchronous source-native steps. P1 moves writes to an owned thread; the intervention includes ownership and control-read changes, with the same payload slices, staging locks and single-buffer limits. A progress witness requires a positive socket write wholly inside an independently verified whole-work `engine.step`, on distinct sender and engine threads. It establishes local socket acceptance during native-step residence, not NIC delivery or exclusive GPU overlap. Complete work, KV readback and ownership are checked. Appendix H specifies the lease, join and installed-acknowledgement protocol and its 21 CPU checks.

Experimental configuration and scope. Both endpoints hold DeepSeek-R1-0528-Qwen3-8B in FP16 with tensor parallelism one (DeepSeek-AI, 2025): Iluvatar BI-V150/vLLM at source and MetaX C500/LMDeploy at receiver. The controlled regime uses one staging slot, one source-prefix lease, fixed routes and complete-output replay of two consumed trajectories. Source execution is eager, prefetch is disabled, and validation and observation remain enabled. Each endpoint initially receives four CPU-equivalents and 48 GiB host memory. Fresh PD validates and installs KV before continuation, checking representation, positions, bytes and ownership (Appendix E).

Workload and targets. Short and long requests have 256- and 3072-token inputs and each emits 195 visible tokens, with 196 replay commits including terminal EOS. A phase alternates these classes at 1.25 requests/s: 20 warmup arrivals over 16 s, followed by 40 measurement arrivals over $H = 32$ s and complete drain. The primary joint target is TTFT at most 2 s, average completion TPOT at most 50 ms and completion at most 30 s. Goodput divides qualifying measurement arrivals by 32 s, including requests that complete later.

Arms and placement. P0 and P1 use a 60-entry vector fixed from an earlier development phase: 24 whole BI, six whole MX and 30 PD actions. The six MX actions occur in warmup; every measured short request executes whole on BI and every measured long request uses PD. W assigns short requests to BI and long requests to MX on the same devices. It was selected from four ordinary controls using the same available input-length and residual-service information and confirmed in a separate A-B-B-A comparison (Appendix F).

Replication and sensitivity. Qualification and reversed W/P0/P1 blocks complete 368 applications per physical allocation. A separate four/sixteen-CPU P0/P1 study adds 512 applications. Primary blocks prespecify native progress and at least 20% long-TTFT improvement; short-success preservation distinguishes benefit from harm. Appendices A–C retain execution orders, qualifications and startup recoveries.

Blocks and uncertainty. Each campaign reports treatment contrasts by independently initialized block. Requests share queues and are not independent replications. We neither pool these requests nor combine blocks from different interventions or arrival conditions into a population confidence interval.

TTFT (s)	TPOT (ms)	W /40	P0 /40	P1 /40	P1-P0
2	50 (primary)	40	20	0	-20
2	60	40	20	20	0
32	60	40	20	40	+20

Table 1: Posthoc rescoring of the original P0/P1 sender experiments, not the matched-ownership experiment. Each row holds separately in all four initialized blocks on the two physical allocations. Completion remains at most 30 s, making the 32 s TTFT limit nonbinding. W is not retuned, and the frozen primary SLO remains unchanged.

3 PHENOMENON: PROGRESS IMPROVES WHILE QUALIFICATION FALLS

In the original blocks, long mean TTFT falls from 46.575/46.690 to 17.019/16.591 s, while short mean TPOT rises from 43.439/42.949 to 55.493/55.490 ms. Both blocks pass the prespecified progress and latency-improvement criteria (Appendix A, Table 2); qualification additionally depends on the joint targets.

Every measured long request still exceeds the 2 s TTFT target, while every P1 short request violates only the 50 ms completion-TPOT target. P1 therefore gains no qualifying long request and loses all 20 short successes, reducing goodput from 0.625 to zero requests/s. W qualifies all 40 measurement requests in both blocks. Figure 4 in Appendix A retains the complete per-request distributions.

3.1 QUALIFICATION TRANSITIONS EXPLAIN THE GOODPUT CHANGE

Let I index the same matched measurement arrivals in both arms, and let $S^a \subseteq I$ be the qualifying requests under arm $a \in \{0, 1\}$, denoting P0 and P1. For a common arrival-window duration H , goodput is $G_a = |S^a|/H$. With newly qualifying requests $R = S^1 \setminus S^0$ and lost successes $L = S^0 \setminus S^1$, shared successes cancel to give

$$G_1 - G_0 = \frac{|R| - |L|}{H}. \quad (1)$$

Goodput does not decrease on this cohort exactly when $|R| \geq |L|$; class protection additionally requires preserving its successes. Here $|R| = 0$, $|L| = 20$ and $\Delta G = -0.625$ requests/s. This offline identity explains the observed change without predicting unobserved arrivals.

Qualification depends on the service targets. Latency changes are observed outcomes; which requests gain or lose qualification depends on the SLO. Rescoring the same historical records yields a loss, a tie or a gain relative to P0 (Table 1). W remains at 40/40 in every listed block and target. These are descriptive re-scorings, not new replications; Appendix D retains the full grid.

4 MECHANISM: AN ADMISSION-DEPENDENT PROGRESS EFFECT

Posthoc traces identify a candidate explanation inside source execution. In both original blocks, the number of BI native steps containing whole-request and prefix query updates rises from 24 to 88. The summed host-observed residence of these mixed steps increases from approximately 4.7 to 17.1 s. Warmup payload reception falls from approximately 3.9 to 0.2 s, and measured long requests spend less time waiting to enter staging (Appendix A). Earlier prefix progression thus co-occurs with substantially different coexistence of the two work classes.

These host traces include synchronization, sampling and scheduling; they do not identify a unique GPU or CPU mediator. The following matched-ownership study tests the progress-admission interaction, while the local-work study tests whether a particular cap restores protection. Together with the arrival and replay tests in Section 5, these are separate campaigns, not a full crossing of all factors.

Algorithm 1 Prefix entry conditioned on current whole-request work

Require: Prefix i ; mode $g \in \{\text{immediate}, \text{empty}\}$; live source state $\mathcal{L}$; prefix lock and remote staging already held

- 1: $d \leftarrow \text{clock}() + 90\text{ s}; \delta \leftarrow 1\text{ ms}$
- 2: **loop**
- 3: If the source has failed, retain the failure receipt and propagate the error
- 4: $\mathcal{U} \leftarrow \{j \in \mathcal{L} : j.\text{role} \neq \text{prefix} \wedge \neg j.\text{execution_complete}\}$
- 5: **if** $g = \text{immediate}$ or $\mathcal{U} = \emptyset$ **then**
- 6: Record admission and $\mathcal{U}$; submit prefix i without yielding; **return**
- 7: **end if**
- 8: If $\text{clock}() \geq d$, retain the timeout receipt and propagate the error
- 9: Yield for δ before observing current state again
- 10: **end loop**

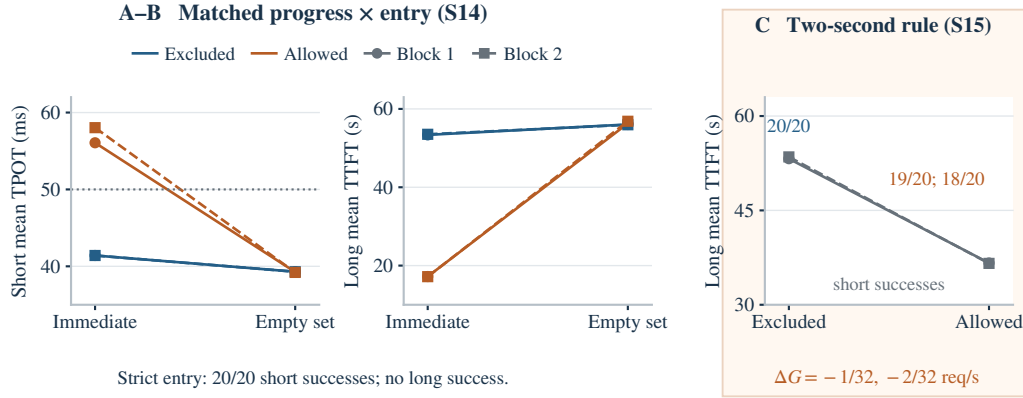


Figure 2: **Protection and latency must be read together.** (A–B) In the matched progress–entry experiment, strict entry restores all twenty short successes while increasing long TTFT; line styles distinguish initialized blocks. (C) A separate fixed two-second experiment retains about 31% lower long TTFT under allowed progress, but loses one/two short successes. The separate background marks a different campaign, not a connected performance frontier. All long requests fail the joint target. Points are block means; success counts use individual requests.

4.1 MATCHED PROGRESS AND PREFIX-ENTRY CONTROLS

Both arms use the same owned sender, descriptor lease, control-read handling, observer and native-step mutex. P0* excludes socket writes from native steps using that mutex; P1* permits them. Each is crossed with immediate entry or waiting for the current whole-request set to empty (Algorithm 1). Both leases are held before the gate, so waiting also reserves staging. P0* differs from historical event-loop P0. Four qualifications precede two opposite-order four-cell blocks on a fresh same-model allocation. The original workload, routes, resources and SLOs remain fixed; all 512 applications pass native and cleanup checks (Appendix I).

4.2 ADMISSION ATTENUATES THE PROGRESS PENALTY AT A COST

Let $Y_{p,g}^{(b)}$ be short mean completion TPOT in block b , progress mode p and gate g . The progress contrast and its admission interaction are

$$D_g^{(b)} = Y_{\text{allow},g}^{(b)} - Y_{\text{exclude},g}^{(b)}, \quad I^{(b)} = D_{\text{empty}}^{(b)} - D_{\text{immediate}}^{(b)}. \quad (2)$$

With immediate entry, D_g is 14.658/16.625 ms in the two blocks: P0* qualifies 20/20 short requests and P1* qualifies none. With empty-set entry, D_g becomes $-0.036/-0.090$ ms, giving interactions of $-14.694/-16.715$ ms. The gate thus attenuates the progress-specific penalty, rather than merely improving both arms equally. Each gated P1* cell rescues all twenty short requests and raises goodput from zero to 0.625 requests/s.

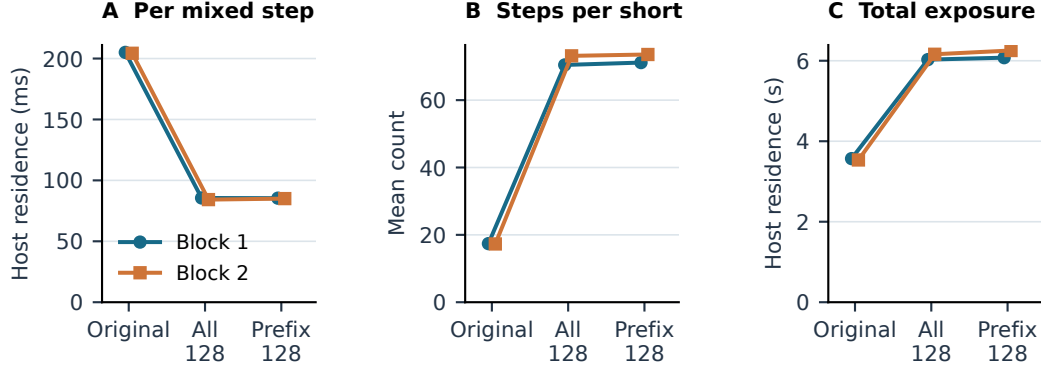


Figure 3: **Smaller native steps can increase cumulative exposure.** Allowed-progress cells from the same native-budget campaign. (A) Mixed-step host residence, weighted over intersections with the source first-visible-to-terminal window of each short request. (B) Mean number of own mixed steps per measured short request. (C) Their mean accumulated host residence. Repeated residence is counted for each affected request; it is not exclusive GPU time. Markers distinguish initialized blocks. All three rules retain immediate admission; C/P use 128-position caps. The figure describes a response to the intervention, not a unique internal mediator.

The protection does not retain the long-TTFT benefit (Figure 2). P1* long mean TTFT rises from 17.208/17.174 to 56.307/56.849 s, exceeding even immediate P0* at 53.374/53.522 s. No long request qualifies. Mixed whole/prefix native steps change from 8/8 to 88/88 when immediate progress is allowed, and are zero under both strict gates. Residual sender/native write witnesses in gated P1* belong only to one warmup-cohort payload per block. The matched experiment establishes an admission-dependent runtime effect; contention timing and a unique GPU mediator remain unresolved.

The resource boundary extends beyond transfer. In supporting A-B-B-A measurements, pre-staging waiting accounts for 96.000% of summed long-request TTFT, while reception averages only 0.189/0.188 s (Appendix F). The staging slot spans prefix creation through installation and acknowledgement. Once that handoff completes and its leases are released, a waiting PD request can acquire staging and issue its source-prefix call; the new prefix remains subject to its lease and admission gate (Figure 1A). This resource dependency is implemented. Whether earlier reuse harms short requests through changed batching, sender CPU competition or other shared work remains only partially isolated.

4.3 SMALLER STEPS CAN INCREASE CUMULATIVE EXPOSURE

A separate 768-application experiment tests a concrete local-work intervention: original work, ordinary 128-position per-request chunking, and a prefix-only 128-position cap, crossed with write exclusion/permission. All cells retain immediate entry and total step budget 1024. Both capped rules preserve measurement-cohort progress but recover no short successes in either block; matched exclusion retains 20/20 throughout.

Figure 3 shows why local step size alone is an insufficient design criterion. With allowed progress, mean windowed mixed-step host residence falls from about 205 to 84–86 ms, but a short request experiences about 70–74 mixed steps rather than 17. Its cumulative mixed-step residence grows from about 3.5 to 6.0–6.3 s. Long mean TTFT also rises from about 17–18 to 35–36 s. The tested caps shorten local work while increasing repeated exposure; they do not restore client protection. Host residence includes shared runtime costs and does not identify exclusive GPU contention. Appendix K gives all cells and the native intervention; this is one cap value, not a complete Sarathi-Serve reimplement.

5 BOUNDARY CONDITIONS

5.1 PHYSICAL REPLICATION

A second allocation of the same accelerator models reproduces the result: long mean TTFT falls from about 46–48 to 17 s, short successes change from 20/20 to 0/20, no long request qualifies, and W remains at 40/40. All four initialized blocks thus show the same qualification loss.

5.2 THE LOSS PERSISTS WITH A WIDER CPU QUOTA

Across four new comparisons at four/sixteen CPU-equivalents, P0 qualifies all 20 short requests. P1 qualifies none at sixteen CPUs and zero or one at four; no long request qualifies. Long mean TTFT improves by 22.734–31.375 s while short mean TPOT rises by 8.985–12.952 ms.

Every phase records zero leaf-cgroup throttling on both hosts, so the loss persists without observed quota exhaustion. Quotas do not reserve cores or eliminate CPU contention. Thread settings stay fixed; W is not rerun (Appendix C).

5.3 A PROSPECTIVE BURST-ARRIVAL BOUNDARY

This separate campaign tests the progress–admission interaction under a new arrival condition. Its vector places four requests 10 ms apart every 3.2 s, retaining the mean arrival rate, work, routes and targets. All 752 applications complete across three initialized four-cell blocks. The short-TPOT interactions are $-13.563/ -12.927/ -12.904$ ms; each block loses twenty short successes with immediate progress and restores them with strict entry. Allowed-progress long TTFT rises from about 20 to 56 s. Strict entry again removes measurement-cohort write witnesses and mixed steps. This extends the interaction to one new arrival condition, with texts still consumed (Appendix L).

5.4 REPLAY PUBLICATION AT THE NATIVE CPU BOUNDARY

This separate immediate-entry campaign tests dependence on replay placement. A 512-application control moves replay from the sampler wrapper to the existing native CPU-result boundary, removing an extra token readback/device roundtrip while retaining native sampling, history, positions and KV checks. The short-TPOT progress penalty remains 16.48/15.52 ms, versus 15.85/16.57 ms under legacy replay; both lose twenty short successes per block. The extra roundtrip is therefore not necessary for this penalty in the tested setting (Appendix M).

6 DESIGN IMPLICATION: CONTROL READINESS AGAINST CLIENT TARGETS

Immediate P1* reproduces the qualification loss with ownership held constant. Strict gating protects short requests while reserving staging and delaying long requests. A useful successor must preserve qualification without serializing away the latency gain; Equation 1 measures the observed tradeoff. Admission itself is established in prior systems (Zhong et al., 2024; Wang et al., 2025; Lin et al., 2026).

A fixed wait threshold tests the tradeoff. A separate, prospectively fixed rule admits a prefix when the whole set empties or gate waiting is observed to reach 2 s, with staging still reserved. All 256 applications complete. Across two opposite-order blocks, allowed progress lowers long mean TTFT by 31.15/31.63% against matched exclusion, but qualifies 19/20 and 18/20 short requests versus 20/20; no long request qualifies. Goodput falls by 1/32 and 2/32 requests/s. Measured-cohort progress remains present. Synchronous native work can delay threshold observation. Figure 2C and Appendix I.1 report this separate campaign; comparisons with earlier gates are descriptive.

The whole-request alternative remains essential. The same pool supports PD and whole-request execution. Supporting matched-replay measurements put PD 6.50–7.59% ahead of fixed whole placement but 3.15–4.59% behind ordinary actual batching (Appendix F). Primary W reaches the finite target’s 40/40 ceiling. A successor algorithm must compete with whole-request execution where the target has measurable headroom.

7 RELATED WORK

DistServe and Splitwise develop PD specialization, while Sarathi-Serve controls prefill/decode co-existence through chunking (Zhong et al., 2024; Patel et al., 2024; Agrawal et al., 2024). TaiChi combines aggregated and disaggregated execution using SLO budgets; FlowPrefill and SLOWeave study preemption and deadline-aware chunking (Wang et al., 2025; Hsieh et al., 2026; Song et al., 2026). The two systems named Kairos address slack-guided batching and TBT-constrained prefill deflection (Wang, 2026; Arun et al., 2026). The latter’s static-deflection baselines trade TTFT gains for TBT violations, providing relevant precedent. Gate-and-route control models state-dependent contention and proves optimality in a specified many-server limit (Lin et al., 2026). Our question isolates existing-payload progress with every actual destination fixed.

Multi-vendor PD, DOPS and HMA-Serve address heterogeneous execution and transfer; KVDirect and FlowKV develop transfer mechanisms and load-aware execution (Chen et al., 2025b; Yang et al., 2026; Wei et al., 2026; Chen et al., 2025a; Li et al., 2025). Revisiting disaggregated serving emphasizes colocated alternatives and conditional benefits (Li et al., 2026); measured routing is sensitive to hardware calibration and pool size (Tumkur et al., 2026). We complement these systems with native sender/admission interventions, fixed placements and replicated client qualification. W is the executed control on the same devices; cited systems establish the design context.

8 SCOPE AND REPRODUCIBILITY

The findings apply to the controlled regime in Section 2: one 8B model, two runtimes and consumed trajectories, serial staging, and the two tested arrival patterns. Physical replication uses the same accelerator models. Finite drained cohorts do not estimate sustainable capacity. Natural-answer quality, energy, broader length distributions, sustained arrivals, multi-slot staging and other architectures remain future work. Removing one replay roundtrip supplies neither an uninstrumented control nor unique mediation.

The companion artifact, [ruxiao/kv-progress-serving-goodput](#) on GitHub (release v42_1), separates record-to-table/figure reconstruction from native rerunning, with versioned code, commands and expected outputs. Its earlier R44/R46 runtime sources match four native qualifications (32 applications); the portable host launcher has only CPU/loopback validation. Vendor images and weights are separate (Appendix E).

The mechanism, bounded-entry, native-budget, arrival-boundary and replay-placement artifacts provide 512/256/768/752/512 application records with independent reconstruction. These campaign projections are not standalone GPU launchers. Archived EoH budgets, failures and controls remain separate (Appendix G).

9 CONCLUSION

Native KV progress interacts with admission: strict entry removes the progress-associated short-TPOT penalty at a long-request latency cost, including under one new microburst pattern. A fixed two-second threshold retains partial latency benefit with incomplete protection. Separately, smaller mixed steps increase cumulative exposure without restoring qualification. The progress effect depends on admission; its individual CPU, GPU, batching and scheduling mediators remain unresolved. The results motivate assessing progress, entry and local work against explicit client targets.

AI INVOLVEMENT AND HUMAN ACCOUNTABILITY

The human authors defined research goals, provided resources, guided priorities and revisions, and made submission decisions. Litlit S1 assisted with literature review, hypothesis development, code implementation, experiment execution, data analysis and manuscript drafting. Image-generation assistance was used for an earlier overview; the present figures are vector drawings of the experimental design and recorded measurements. DeepSeek-R1-0528-Qwen3-8B generated earlier EoH candidates; a separate archived search used Qwen3-14B. The authors retain responsibility for verifying the final manuscript and its scientific claims, data rights, disclosures and submission.

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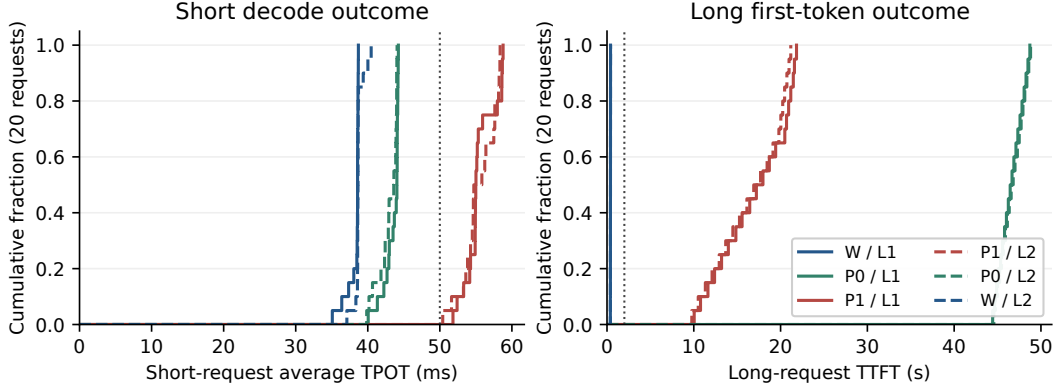


Figure 4: All 20 measurement requests per class and arm on the original allocation. Solid/dashed lines distinguish initialized blocks; vertical lines show the frozen SLOs. These are complete cohort distributions, rather than population confidence intervals.

A FIXED-PLACEMENT SENDER COMPARISON

The main experiment fixes the complete 60-entry route vector from an earlier development run: 24 whole BI, six whole MX and 30 PD actions. The six MX requests occur in warmup; every measured short request runs whole on BI and every measured long request uses PD. The vector is consumed development information used as a controlled assignment, rather than an online policy. Each executed destination is checked against the frozen plan and decision digest. Timing, batching and queue state can respond to the sender intervention while destinations remain fixed.

Table 2: Original allocation in executed order. Each arm has 20 measured requests per class after 20 warmup arrivals. Latencies are class means; goodput uses the common 32 s arrival window.

Block	Arm	Short /20	Long /20	Goodput req/s	Long TTFT s	Short TPOT ms
L1	W	20	20	1.25000	0.397	38.224
L1	P0	20	0	0.62500	46.575	43.439
L1	P1	0	0	0.00000	17.019	55.493
L2	P1	0	0	0.00000	16.591	55.490
L2	P0	20	0	0.62500	46.690	42.949
L2	W	20	20	1.25000	0.398	38.737

Protocol and primary contrast. Eight qualification applications precede two independently initialized 180-application blocks in W/P0/P1 and P1/P0/W order. Both performance plans are frozen before any performance output. Each arm has 20 warmup and 40 measurement arrivals at 1.25 requests/s, followed by full drain. The matched replay, model, precision, device pool, quotas and SLOs are identical. Qualification checks correctness and sender progress; it does not select favorable performance. All 368 applications complete once.

The prespecified harm criterion requires actual sender progress, at least 20% lower mean long-request TTFT, and fewer qualifying short requests in both blocks. The earlier engineering-benefit criterion instead requires short-success preservation. These distinguish an identified client effect from an improvement in serving. Exact integer sums test the TTFT criterion; correlated requests do not supply independent uncertainty estimates. Figure 4 includes every measured request.

Executable comparison procedure. Algorithm 2 specifies the matched evaluation of the historical P0/P1 sender bundle and its separate W control.

Native evidence and work. We match complete application trajectories using output replay. After each native sampling operation, a fixed reference token is committed as subsequent history. This

Algorithm 2 Fixed-route evaluation of a PD progress intervention

Require: Frozen arrivals A (including warmup), routes ρ , replay work Ξ , resources C , SLOs τ , window H , opposite arm orders

Ensure: Per-block goodput change and identities of rescued/lost requests

```

1: for each independently initialized block  $b$  do
2:   for each arm  $a \in \{P0, P1, W\}$  in the frozen order do
3:     Execute  $A$  with  $\Xi, C$ ; use  $\rho$  for P0/P1 and W's fixed placement
4:     Drain all work; retain native receipts and one-clock client events
5:     Check routes, work, KV validity; for P1, verify native progress
6:     If evidence is incomplete, retain the failed run and stop the comparison
7:      $S_b^a \leftarrow \{i \text{ in measurement cohort} : i \text{ meets all SLOs } \tau\}$ 
8:   end for
9:    $R_b \leftarrow S_b^{P1} \setminus S_b^{P0}; L_b \leftarrow S_b^{P0} \setminus S_b^{P1}$ 
10:  Report  $G_b^a = |S_b^a|/H, \Delta G_b = (|R_b| - |L_b|)/H$ , class counts
11: end for

```

makes both arms execute the same application token trajectory while preserving native queries, sampling and the effect of their scheduling. The accounting distinguishes committed samples, source-prefix work, discarded partial-prefill choices and client-visible tokens. Thus differences in native batch composition remain observable, rather than being mistaken for differences in the requested application work. Replay is an experimental control for cost and timing; natural-answer quality is outside its estimand.

The routing loop records receipt publication after its dictionary update and records decision observation without an intervening await. An independent checker reconstructs publication order and the corresponding prefix-pending flag. Actual destinations are also bound independently. This validates the fields needed for the fixed-route comparison; it does not reconstruct every dynamic queue or lock field.

The original allocation commits 72,128 application samples and emits 71,760 visible tokens. There are additionally 122 source-prefix samples and 305 discarded partial-prefill choices, giving 72,555 native sampler invocations and 684,112 scheduled query positions. Each performance arm commits 11,760 application samples and schedules 111,540 query positions. P0 and P1 each add 30 source-prefix samples; discarded partial-prefill choices differ, 66 versus 82 per block. Matching complete application trajectories therefore retains observable differences in native scheduling overhead.

Table 3: All actual mixed BI steps and all ten warmup PD arrivals in each fixed-placement arm. Step residence is host elapsed time including synchronization, sampling and scheduling.

Block	Arm	Mixed steps	Caller wall (s)	Warmup receive (s)
L1	P0	24	4.743	3.906
L1	P1	88	17.114	0.192
L2	P1	88	17.151	0.196
L2	P0	24	4.661	3.939

Posthoc diagnostic. Caller wall time is not exclusive GPU service; receive time includes runtime scheduling. These associations identify no unique mediator.

Posthoc mixed-step membership is checked against native query increments. Summed residence counts each step once. The changed interleaving in Table 3 is an observed response to the sender bundle; it does not isolate a unique GPU or CPU mediator. The bundle includes ownership and control-read changes as well as payload writes.

The reconstruction uses anonymous client, work, route and publication projections. Two collection-checker corrections—canonicalizing tuple/list scores and opening a root-owned lock read-only—occur without additional GPU applications or altered measurements. Their failed records remain archived. Runtime sources and raw-tensor checks support native validity separately from arithmetic reconstruction.

B REPLICATION ON ANOTHER PHYSICAL ALLOCATION

The second allocation uses a BI-V150 in another host and another physical C500, with the same accelerator models, weights, images, precision, trajectories, routes and quotas. Eight qualification applications precede the same opposite-order blocks; both performance plans are frozen together. All 368 applications complete once, including 120 performance warmup and 240 measurement arrivals. This is hardware-allocation replication on the consumed workload.

Table 4: Alternate allocation in executed order. Each arm has 20 measured requests per class and 20 warmup arrivals. Latencies are class means; qualification uses each request’s joint SLO.

Block	Arm	Short /20	Long /20	Goodput req/s	Long TTFT s	Short TPOT ms
L1	W	20	20	1.25000	0.397	38.393
L1	P0	20	0	0.62500	47.711	42.891
L1	P1	0	0	0.00000	17.051	55.467
L2	P1	0	0	0.00000	17.006	55.174
L2	P0	20	0	0.62500	46.161	43.528
L2	W	20	20	1.25000	0.397	38.767

On the alternate physical pair of the same two device SKUs, with the same 60-request action vector in P0 and P1, L1: long mean TTFT 47.711 to 17.051 s, short mean TPOT 42.891 to 55.467 ms, and short joint-SLO success 20/20 to 0/20; L2: long mean TTFT 46.161 to 17.006 s, short mean TPOT 43.528 to 55.174 ms, and short joint-SLO success 20/20 to 0/20. Both initialized blocks meet the prespecified fixed-route tradeoff criterion: at least 20% lower long mean TTFT with observed sender progress, accompanied by fewer short joint-SLO successes. Each arm retains all 40 measurement arrivals (20 short and 20 long), with the matched ordinary whole-placement reference shown separately; it was not screened as the strongest baseline on this pair. The vector was selected from consumed development data; these are bounded technical replications, not independent workload populations. The experiment isolates the sender implementation at fixed placement, not a unique internal CPU, NIC or GPU submechanism. All failed P1 short requests violate only average completion TPOT; every long request still violates TTFT. Work totals are 72,128 application commits, 71,760 visible tokens, 122 source-prefix samples, 304 discarded partial-prefill choices, 72,554 native sampler invocations and 684,112 scheduled query positions.

Preparation accounting. One L2 preparation attempt stopped before pair commit, model launch, or request execution after host image garbage collection removed the frozen BI image. The exact image was restored and held by a temporary sleeping asset-reference container with no GPU, model mounts, or network; that owned helper was removed after L2. The same frozen plan and source then ran once, with zero application reruns. Original failed preparation artifacts and all recovery receipts are retained by hash. The sleeping helper was outside the experimental containers, capped at 0.05 CPU and 16 MiB memory, and present throughout all three second-block arms. Its overhead was not separately measured. No performance outcome selected a retry.

C CPU-QUOTA SENSITIVITY AND CAMPAIGN ACCOUNTING

On the second allocation, each endpoint receives either four or sixteen CPU-equivalents, one accelerator and 48 GiB host memory. OpenMP/OpenBLAS thread settings remain four/one, with unchanged affinity sets of 255 logical CPUs at BI and 64 at MX. These are eligible CPUs, not reserved cores. Four eight-application qualifications and four 120-application plans are frozen before new native output. Orders are 4/P0–P1, 16/P0–P1, 16/P1–P0 and 4/P1–P0. All 512 applications complete once: 32 qualification, 160 performance warmup and 320 measurement.

All four qualification runs (32 applications) and four performance lifecycles (480 applications; 320 measured and 160 warmup) passed the integrity audits. At the configured 4-CPU quota, P1 minus P0 long mean TTFT differences are -22.734 s and -31.375 s, and short joint-SLO count differences are -20 and -19. At the configured 16-CPU quota, P1 minus P0 long mean TTFT differences are -30.963 s and -29.751 s, and short joint-SLO count differences are -20 and -20. The table retains

Table 5: All CPU-sensitivity arms in executed order. Short/long counts are out of 20 measurement requests each; goodput is requests/s, long TTFT seconds and short mean completion TPOT milliseconds.

Block (CPU)	Arm	Short	Long	Goodput	Long TTFT	Short TPOT
L1 (4)	P0	20/20	0/20	0.625	40.239	46.006
L1 (4)	P1	0/20	0/20	0.000	17.505	54.990
L2 (16)	P0	20/20	0/20	0.625	47.971	42.753
L2 (16)	P1	0/20	0/20	0.000	17.008	55.207
L3 (16)	P1	0/20	0/20	0.000	16.698	55.191
L3 (16)	P0	20/20	0/20	0.625	46.449	43.999
L4 (4)	P1	1/20	0/20	0.031	16.159	54.378
L4 (4)	P0	20/20	0/20	0.625	47.535	41.426

every quota/order condition; signs are descriptive order-matched contrasts, with no independent-request confidence interval. CPU counters cover whole phases including warmup and drain, and do not isolate GPU service or an exclusive causal mediator. P1 demonstrates positive-write progress in every performance lifecycle; P0 has none. The reverse-order four-CPU P1 arm retains one short success. Long requests miss the target in every arm. Order-matched differences between the 16- and 4-CPU treatment effects have mixed signs: $-8.229/+1.624$ s for long TTFT and $+3.470/-1.761$ ms for short TPOT. This supports persistence of the qualification loss, rather than a consistent quota interaction.

Both hosts record zero leaf-cgroup throttled-period and throttled-time increments in all eight performance phases. The sampled intervals include warmup, measurement, drain and cleanup. Other host contention and ancestor constraints remain possible; quota changes neither reserve cores nor identify a unique processor mediator.

Complete failure accounting. An earlier campaign (R45) completed 32 qualification applications before its first performance lifecycle failed native plan validation. That failed lifecycle had 0 actual applications, for 32 total applications in the stopped campaign; it yields no successful performance estimate. Source-side vendor initialization did execute; zero offered applications does not mean zero GPU initialization. The remaining three performance lifecycles were not started. R46 uses a separately frozen corrected source and repeats all four qualification and four performance lifecycles. The stopped predecessor accepted only one- or three-phase receiver plans, rejecting its two-arm performance plan before any client application. The corrected source validates the two-arm contract and enables its counter hooks, while preserving sender computation, work and SLOs. The public kit binds 30 runtime modules to the successful qualification deployment; its portable host launcher has CPU/loopback checks but awaits GPU execution. A GPU-free sleeping image-retention helper with the same 0.05-CPU/16-MiB cap is present throughout these campaigns, then removed. Initial failures, source origins and retained services are recorded separately.

D POSTHOC JOINT-SLO SENSITIVITY

All 480 measurement arrivals from the four primary blocks are reused. Integer client events and exact rational TPOT first reproduce the primary counts. The complete chosen grid is TTFT seconds $\{0.5, 1, 2, 4, 8, 16, 32, 64\}$ crossed with TPOT milliseconds $\{20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 80\}$, retaining completion ≤ 30 s and the 32 s arrival denominator. Its 96 cells per block are posthoc summaries of the same observations, rather than independent tests.

The largest P0-short TPOT is 44.2464452577 ms and the smallest P1-short TPOT is 50.1267871495 ms; all these requests meet the 2 s/30 s conditions. Their lower-inclusive, upper-exclusive interval therefore gives complete short-success loss, including the primary 50 ms target. Full-precision events determine counts before display rounding.

Observed target dependence. The same recorded intervention can harm, preserve or improve goodput under different targets. We examine this dependence descriptively by recomputing all 480

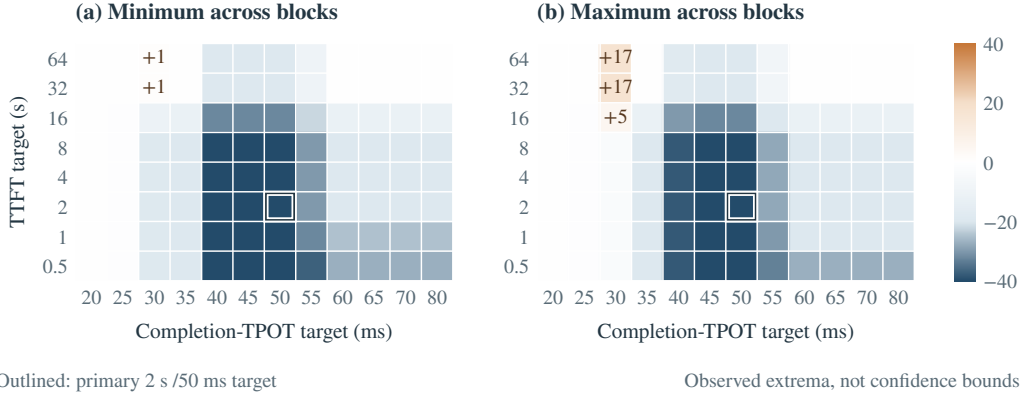


Figure 5: **SLO dependence of the observed serving comparison.** Minimum and maximum P1-minus-W success counts across four blocks at each target; these are observed extrema, not confidence bounds. The box marks the primary 2 s/50 ms target. Positive cells are annotated. Completion is fixed at 30 s, so the 32 s and 64 s TTFT rows coincide. All 384 cell-block comparisons reuse the same 480 measurement applications.

measurement applications in the four primary blocks from integer client events, using exact rational TPOT. The complete posthoc grid contains eight TTFT and twelve TPOT thresholds at a fixed 30 s completion target (Appendix D).

At TTFT 2 s, all P0 short requests qualify and all P1 short requests fail throughout the common TPOT interval $[44.246445, 50.126787)$ ms, with rounded displayed endpoints. Both 45 and 50 ms therefore produce the full loss. At 55 ms, P1 qualifies 10, 10, 11 and 12 requests across the four blocks; at 60 ms, P0 and P1 tie at 20/40. With TTFT relaxed to 32 s and TPOT to 60 ms, P1 qualifies 40/40 against P0’s 20/40: the same accounting now gives 20 rescued and zero lost requests per block. Completion within 30 s remains required.

The comparison with W also changes across targets. At TTFT 32 s and TPOT 30 ms, P1 qualifies 20/40 in each block, whereas W qualifies 14/40, 3/40, 19/40 and 3/40. Figure 5 includes the full grid, including these favorable P1 cells. W is the control selected at the primary target and is not retuned for this sweep. These posthoc results characterize the observed trajectories’ SLO dependence; the primary frozen target and its outcome remain unchanged.

TTFT (s)	TPOT (ms)	W /40	P0 /40	P1 /40 (four blocks)
2	45	40	20	0, 0, 0, 0
2	50 (primary)	40	20	0, 0, 0, 0
2	55	40	20	10, 10, 11, 12
2	60	40	20	20, 20, 20, 20
32	60	40	20	40, 40, 40, 40

Table 6: Illustrative cells from the full posthoc grid. Completion remains at most 30 s. Counts are shown in original L1/L2 then alternate-allocation L1/L2 order; W and P0 counts are identical across these blocks at these cells. Goodput is each count divided by 32 s. At 32 s TTFT, completion is the tighter total-time condition. Relaxed targets do not replace the primary SLO.

The following tables retain every grid cell as the difference in qualified requests, P1 minus P0. Full per-arm counts and goodput, including W, are exported in `generated/slo_posthoc_summary.json`. Negative, zero and positive differences delimit harm, ties and improvement relative to P0 within the same data. Improvement over P0 is not superiority over W, and the grid does not estimate the prevalence of such regimes in a workload population.

TTFT (s) / TPOT (ms)	20	25	30	35	40	45	50	55	60	65	70	80
0.5	0	0	0	0	0	-18	-18	-11	-5	-5	-5	-5
1	0	0	0	0	0	-19	-19	-11	-2	-2	-2	-2
2	0	0	0	0	0	-20	-20	-10	0	0	0	0
4	0	0	0	0	0	-20	-20	-10	0	0	0	0
8	0	0	0	0	0	-20	-20	-10	0	0	0	0
16	0	0	8	8	8	-12	-12	-2	8	8	8	8
32	0	1	20	20	20	0	0	10	20	20	20	20
64	0	1	20	20	20	0	0	10	20	20	20	20

Table 7: Original allocation, block L1: all 96 posthoc cells as P1-minus-P0 qualified count out of 40. Completion is bounded by 30 s throughout.

TTFT (s) / TPOT (ms)	20	25	30	35	40	45	50	55	60	65	70	80
0.5	0	0	0	0	0	-17	-17	-14	-4	-4	-4	-4
1	0	0	0	0	-1	-20	-20	-10	0	0	0	0
2	0	0	0	0	-1	-20	-20	-10	0	0	0	0
4	0	0	0	0	-1	-20	-20	-10	0	0	0	0
8	0	0	0	0	-1	-20	-20	-10	0	0	0	0
16	0	0	8	8	7	-12	-12	-2	8	8	8	8
32	0	1	20	20	19	0	0	10	20	20	20	20
64	0	1	20	20	19	0	0	10	20	20	20	20

Table 8: Original allocation, block L2: all 96 posthoc cells as P1-minus-P0 qualified count out of 40. Completion is bounded by 30 s throughout.

TTFT (s) / TPOT (ms)	20	25	30	35	40	45	50	55	60	65	70	80
0.5	0	0	0	0	0	-18	-18	-11	-5	-5	-5	-5
1	0	0	0	0	0	-19	-19	-11	-4	-4	-4	-4
2	0	0	0	0	0	-20	-20	-9	0	0	0	0
4	0	0	0	0	0	-20	-20	-9	0	0	0	0
8	0	0	0	0	0	-20	-20	-9	0	0	0	0
16	0	0	8	8	8	-12	-12	-1	8	8	8	8
32	0	1	20	20	20	0	0	11	20	20	20	20
64	0	1	20	20	20	0	0	11	20	20	20	20

Table 9: Alternate same-model allocation, block L1: all 96 posthoc cells as P1-minus-P0 qualified count out of 40. Completion is bounded by 30 s throughout.

TTFT (s) / TPOT (ms)	20	25	30	35	40	45	50	55	60	65	70	80
0.5	0	0	0	0	0	-18	-18	-11	-5	-5	-5	-5
1	0	0	0	0	0	-19	-19	-10	-4	-4	-4	-4
2	0	0	0	0	0	-20	-20	-8	0	0	0	0
4	0	0	0	0	0	-20	-20	-8	0	0	0	0
8	0	0	0	0	0	-20	-20	-8	0	0	0	0
16	0	0	8	8	8	-12	-12	0	8	8	8	8
32	0	1	20	20	20	0	0	12	20	20	20	20
64	0	1	20	20	20	0	0	12	20	20	20	20

Table 10: Alternate same-model allocation, block L2: all 96 posthoc cells as P1-minus-P0 qualified count out of 40. Completion is bounded by 30 s throughout.

E CONFIGURATION AND EXECUTABLE MEASUREMENT METHOD

The bridge uses TCP and host staging. The tested numerical reference reconciles a rotary-amplitude mismatch at amplitude 1.0; raw-tensor checks remain distinct from anonymous event reconstruction. Both engines run eager execution with a 4096-position limit and 16-position cache blocks; prefix caching and prefetch are disabled. BI permits eight sequences and 1024 batched tokens with chunked prefill. MX permits eight sequences, 2048 prefill tokens and a prefill interval of one. These are the frozen experimental settings. The private handoff pins images and vendor-source fingerprints; the anonymous artifact provides complete input/token trajectories and reconstruction code.

Applying the method requires an application-to-native join: every scheduled query carries its member identity, history extent and query positions, and its sampled/committed/visible tokens and termination join to client emissions. One local timeline records stage entry, receipt, publication, validation, installation and release. Cross-host joins use identities and content hashes; unsynchronized absolute clocks are never subtracted. Executed destinations are checked separately from policy scores.

These observations enable three useful comparisons. Complete-work matching distinguishes native scheduling changes from reduced application work. Fixed destinations evaluate a progress intervention without route feedback. Request-level SLO sets quantify whether gains compensate for lost successes. An adopter additionally executes available whole-request batching controls and checks positive transport progress during native work. The scripts implement these tests for the exported schema; runtime adaptation still requires independent validation of its hooks.

F SUPPORTING CONTROLS AND HANDOFF BOUNDARIES

An earlier selected cohort motivated the whole-request comparator. Before matched replay, PD mean completion was 5.314 s versus 6.099 s for fixed whole placement, but their sample counts differed, 428 versus 460. Matching replay fixes 196 background and 264 foreground commits including EOS, 1025-token inputs and a 200 ms foreground arrival offset. P runs background whole MX and foreground BI-to-MX PD; V runs background whole BI and foreground whole MX; R runs both whole MX with actual shared batching. Every route commits 460 samples and schedules 2508 query positions, including prefix work.

Table 11: Matched-work selected cohort: mean completion in seconds. All routes use the same available device pool and complete application work.

Order	P (s)	V (s)	R (s)	Δ_V	Δ_R
PVR	5.692	6.160	5.442	-7.59%	+4.59%
RVP	5.686	6.081	5.512	-6.50%	+3.15%

P beats V by 6.50–7.59% but loses to R by 3.15–4.59% in reversed orders. Actual admission, rather than a configured batch flag, establishes the control: raising the configured maximum from two to three enables two-member batches under the pinned scheduler. All six executions retain 12 applications, 2760 commits, two source samples, two discarded choices and 15,048 query positions. This is a selected-cohort comparison, not an optimality claim.

At 1.25 requests/s, four ordinary controls receive the same arrived input length and residual-service information. All-MX, short-MX/long-BI, the earlier whole router and short-BI/long-MX qualify 0, 20, 9 and 40 of 40 requests. The last becomes W. A separately frozen A-B-B-A comparison confirms W at 40/40 and long-PD at 20/40 twice, retaining all 240 applications. This closes that finite target’s improvement headroom.

All 60 PD handoffs are partitioned using planned arrival, dispatch, staging entry, first read, last byte, validation entry/exit, installation entry/exit and first client emission. Nine consecutive nonnegative intervals telescope to TTFT on one clock. Across 40 measured long requests, pre-staging waiting contributes 96.000% of summed TTFT, while reception averages 0.189/0.188 s. Warmup PD reception is 3.927/3.920 s and stage-through-ACK occupancy 5.777/5.775 s; this work continues into the

Table 12: A-B-B-A long-PD timings in seconds. Cohorts follow planned arrival; selected components do not sum to TTFT. All nine consecutive intervals are exported.

Arrival cohort	B1 warm	B1 meas.	B2 warm	B2 meas.
Requests	10	20	10	20
TTFT (s)	27.441	46.099	27.770	46.046
Before stage (s)	21.651	44.260	21.982	44.199
Prefix setup (s)	1.269	1.063	1.268	1.072
Socket receive (s)	3.927	0.189	3.920	0.188
Validation (s)	0.442	0.442	0.444	0.442
Hold to ACK (s)	5.777	1.802	5.775	1.804

measurement window. ACK precedes protected close/join and unlock, so these occupancy intervals give lower bounds rather than exact resource release.

The same 240 applications contain 180 whole and 60 split requests, with 300 native sequences. Reconstructed contiguous queries total 111,540 positions per phase: a long whole request uses $3072 + 195 = 3267$ and its split counterpart $3071 + 196 = 3267$; a short whole request uses $256 + 195 = 451$. These counts bind work, while timing remains context- and batch-dependent.

G DEVELOPMENT HISTORY AND SCOPE OF CONFIRMATION

The primary study was preceded by several distinct development campaigns. Their original data, scripts, failures and detailed provenance remain in the artifact and archived full research manuscript. They are summarized here to preserve selection history without conflating their populations with the fixed-route comparison.

Dynamic sender experiment. An eight-application qualification preceded two 180-application dynamic-policy lifecycles. Mean long TTFT fell from 45.962/45.922 s to 17.433/16.655 s; short successes fell from 20/20 to 5/20 and 6/20. All long requests still failed TTFT; W reached 40/40. Actual short destinations changed. One second-lifecycle publication discrepancy was action-sensitive, and its independent publication instant was absent; the policy-publication audit consequently remains failed despite complete client/work accounting. No unexecuted route was assigned a counterfactual latency. This motivated freezing every route and recording publication on the routing loop in the main experiment. Collection corrections and failed checks are retained without native retries.

Workloads and streaming controls. Natural-answer confirmation did not reproduce the proposed ranking: an independent workload failed qualification, and a newly qualified concise workload failed the ranking prediction. Earlier streaming evidence includes 158 offered/156 completed applications out of 160 planned and possible external transfer during its second block. Later qualified comparisons found that ordinary chunking removed a claimed unique PD-feasibility advantage. These outcomes constrain generalization beyond the two consumed trajectories; they are not confirmations of a scheduling mechanism.

Load and receiver development. Ordered load curves cover 1080 event-recorded applications, including 360 warmup and 720 measurement arrivals. At corrected-receiver rates 0.125, 0.25 and 0.5 requests/s, PD qualifies all measured requests; at 1 request/s it qualifies 8/32, compared with 16/32 for the earlier receiver. Whole placement qualifies all requests at each rate. The ordered development curves do not isolate a receiver treatment effect. An unresolved lock flag limits resource-availability inference; it does not replace the independently recorded client outcomes.

Heuristic-search lineage. Earlier three- and six-arrival native searches, measured-cost proxy studies and matched-lifecycle searches exposed comparator, state-publication and evaluation defects or failed confirmation. A later EoH study (Liu et al., 2024) used a separate block-output contract. Four model calls produced two natively executed policies and two proposals rejected by legality checks; the executed policies qualified 7/32 and 10/32 requests, versus 21/32 for the strongest ordinary controls. Neither opened independent confirmation. The source artifact retains the complete

study in `event_search_appendix.tex`, including budgets, programs, native trials and failures; `recompute_sl3.py` reconstructs the final comparison. These development records establish no evolved algorithm advantage. The paper’s contribution rests on the qualified fixed-placement finding, runtime evidence and measurement procedure.

H SENDER IMPLEMENTATION AND CPU PROTOCOL CHECKS

The historical sender comparison evaluates an executable bundle: P1 retains the existing descriptor, payload slices, prefix and staging locks, one buffer per endpoint, and KV validation while changing payload execution, ownership and control-read behavior. A positive socket write wholly inside a verified whole-work native step is the progress witness. The owned thread exits before the borrowed descriptor or payload is released; PD continuation requires the installed acknowledgement. Complete work, imported KV readback and receipt publication are checked independently (Appendix A).

Algorithm 3 P1: payload sending with an exclusive descriptor lease

Require: Caller owns TCP writer w and live payload view B ; timeout D

Ensure: Return full-send receipt, or propagate failure after sender cleanup

- 1: Check plain, open, nonblocking socket; empty write buffer; active reads
- 2: Pause transport reads; recheck socket identity and empty write buffer
- 3: Start one owned sender thread t with stop flag $c \leftarrow \text{false}$

Sender thread:

- 4: $k \leftarrow 0$
- 5: **while** $k < |B|$ **do**
- 6: Fail on c , expired D , or exhausted progress-record budget
- 7: Write from offset k up to the next 1 MiB boundary
- 8: On interruption retry; on would-block wait for writability and retry
- 9: Require positive byte count n ; record write interval; $k \leftarrow k + n$
- 10: **end while**

- 11: Record completion or error, then exit t

Event loop:

- 12: Await t while other native work can run; on failure/cancellation set c
 - 13: **Finally:** join t if started, despite cancellation; resume paused reads if open
 - 14: Propagate cancellation/error; otherwise require $k = |B|$ and return receipt
-

The probe calls the unchanged native `r42_sender.py`. Three frozen payload sizes—65,536, 1,048,593 and 4,194,304 bytes—each execute seven conditions once. Fault injection changes only the sender’s OS-write hook; real loopback TCP returns a synthetic acknowledgement after receiving the entire payload. No GPU model or KV installation is involved.

Table 13: CPU protocol cases at three payload sizes. A passed fault case means the expected failure and ownership cleanup occurred.

Condition	Expected outcome	Passed / cases
P0 event-loop send	Payload and synthetic ACK	3/3
P1 owned send	Payload and synthetic ACK	3/3
P1 partial + EINTR/EAGAIN	Payload and synthetic ACK	3/3
P1 zero-byte write	Error, joined cleanup	3/3
P1 injected broken pipe	Error, joined cleanup	3/3
P1 deadline expiry	Error, joined cleanup	3/3
P1 cancellation	Cancellation, joined cleanup	3/3

All 21 cases pass: nine reproduce exact payload hashes and acknowledgements, while twelve propagate failure without a success receipt. Every P1 thread exits before join completion and read resumption; descriptor identity and nonblocking status remain unchanged. Partial-write cases cap writes at 4093 bytes. Each cancellation case exercises one cancellation; repeated cancellation is not separately measured. These checks support the exclusive-caller protocol under the tested faults. They do not certify arbitrary concurrent descriptor use or fault-tolerant native KV recovery.

I MATCHED NATIVE PROGRESS AND PREFIX-ENTRY INTERVENTION

The mechanism study uses the same 8B weights, FP16 precision, eager vendor runtimes, input/replay trajectories, fixed 60-route vector and primary client SLOs on a fresh BI-V150/C500 allocation. Each endpoint retains one accelerator, four CPU-equivalents and 48 GiB host memory. $P0^*/P1^*$ both use the same owned descriptor, sender thread, control-read pause/resume, joins, observer and native-step mutex. $P0^*$ acquires this mutex for each nonblocking write, excluding positive write/native-step intersections; $P1^*$ omits write exclusion. Native steps acquire the common mutex in both arms. This comparison differs from historical $P0/P1$ and contains no newly executed W control.

Admission, order and coverage. The entry factor either admits immediately or waits until the source’s currently observed unfinished whole-request set is empty, including whole prefill and decode. The source prefix lock and gateway staging lease already exist. No native mutex is held during admission waiting, and later whole arrivals may still coexist with an admitted prefix. The strict gate polls cooperatively every 1 ms with a 90 s wait timeout; the sender, acknowledgement and failure-cleanup protocol remain unchanged.

Four eight-application qualifications complete before performance. Block 1 executes excluded/immediate, allowed/immediate, excluded/empty and allowed/empty; block 2 reverses this order. Every performance cell has 20 warmup and 40 measurement requests at 1.25 requests/s, the same 32 s arrival denominator and full drain. All 12 initialized runs pass native/client coverage, complete replay, route, KV and cleanup checks: 512 applications comprise 32 qualification, 160 warmup and 320 measurement applications. Every cell has 20 measured requests per class. Results and interactions are reported per initialized block, without request-independent confidence intervals.

Table 14: All mechanism performance cells in executed order. Short/long successes are each out of 20; TPOT is short mean completion TPOT and TTFT is the long mean. Mixed steps contain actual whole and prefix query updates across the entire initialized phase.

Block	Progress	Entry	Short/long	TPOT (ms)	TTFT (s)	Mixed
1	P_0^*	Immediate	20/0	41.409	53.374	8
1	P_1^*	Immediate	0/0	56.068	17.208	88
1	P_0^*	Empty	20/0	39.296	56.029	0
1	P_1^*	Empty	20/0	39.261	56.307	0
2	P_1^*	Empty	20/0	39.206	56.849	0
2	P_0^*	Empty	20/0	39.295	55.952	0
2	P_1^*	Immediate	0/0	58.033	17.174	88
2	P_0^*	Immediate	20/0	41.407	53.522	8

The within-block progress contrasts under immediate entry are 14.658411 and 16.625267 ms; under empty-set entry they are -0.035614 and -0.089574 ms. Equation 2 therefore gives -14.694026 and -16.714842 ms. Strict entry rescues all twenty short successes of $P1^*$ in each block without losing another measured success, but its long TTFT is 2.933725/3.327181 s worse than immediate $P0^*$. It retains none of the corresponding immediate-progress long-TTFT gain. All long requests remain outside the joint target.

Progress and coexistence observations. Every excluded performance cell has zero positive-write/native-step intersections. Allowed/immediate cells contain 14,232/12,475 strict positive-write witnesses across 22/23 payloads. Empty-set $P1^*$ has 546/639 witnesses, each confined to one warmup payload (the fourth warmup arrival). Warmup denotes the arrival cohort, not an execution interval guaranteed to end before measurement. These residual witnesses do not establish retained measurement-cohort sending overlap. Empty-set admission has zero mixed whole/prefix native steps in both progress arms; the distinction between mixed native execution and sender/native overlap is essential.

Immediate $P1^*$ uses less aggregate sender CPU than $P0^*$ in block 1 (3921.3 versus 4786.3 ms), but slightly more in block 2 (4741.5 versus 4729.3 ms). The short-TPOT penalty occurs despite opposite changes in aggregate sender CPU across blocks; these totals cannot isolate CPU contention timing.

Table 15: Entire-phase mechanism telemetry, including warmup and drain. Payloads count those with a strict whole-step write witness; CPU is the sum of within-thread sender CPU deltas. Gate waiting is averaged across all 30 PD payloads. These quantities do not identify GPU execution time or a unique mediator.

Block	Progress	Entry	Payloads	Writes	CPU (ms)	Gate wait (ms)
1	P_0^*	Immediate	0	0	4786.3	0.0
1	P_1^*	Immediate	22	14232	3921.3	0.0
1	P_0^*	Empty	0	0	4703.7	1117.9
1	P_1^*	Empty	1	546	4468.2	1628.5
2	P_1^*	Empty	1	639	4361.0	1622.8
2	P_0^*	Empty	0	0	4674.1	1116.8
2	P_1^*	Immediate	23	12475	4741.5	0.0
2	P_0^*	Immediate	0	0	4729.3	0.0

Native mixed-step residence is host elapsed time including synchronization, sampling and scheduling. The factorial interaction supports an admission-dependent runtime effect, not exclusive GPU mediation.

Preparation and reproducibility. A predecessor preparation attempt stopped before commit or model execution because an unrelated stale filesystem-handle warning was treated as a host-preflight failure; it offered zero applications. That failure remains archived separately. The completed campaign preserves the frozen scientific factors and workload while correcting warning handling and checking idle device status. No failed preparation is counted as performance or relabeled as a completed application.

The anonymous `mechanism_artifact` supplies all 512 integer client trajectories, the scientific plan, phase-mechanism summaries, source SHA references and exact copies of the three executed control modules. Its standard-library `summarize.py` independently reconstructs client outcomes, class means and interactions. Qualification and warmup are separated from performance counts. Private deployment records, runtime dependencies, images, weights and raw KV tensors are not replaced by this projection. The existing `nativekit` remains the R46 artifact and retains its previously stated portable-launcher boundary.

I.1 FIXED OBSERVED TWO-SECOND ENTRY

The extension preserves the matched owned-sender implementation, routes, workload, resources and client targets, replacing strict waiting with one threshold fixed before its native results. From the start of gate waiting, admission occurs at the first observation of an empty unfinished-whole set or elapsed time of at least 2 s. Empty takes precedence when both conditions hold. Each loop checks source failure and the 90 s safety timeout before eligibility, polls cooperatively every 1 ms, and submits after a successful observation without yielding. Source prefix and remote staging leases remain held. The request’s arrival, SLO and sender/receiver deadlines do not change. This is an observed threshold, not a hard wall-clock bound.

Two eight-application qualifications precede performance. Block 1 executes excluded then allowed progress; block 2 reverses the order. All six runs complete once with native and cleanup checks passing: 256 applications comprise 16 qualification, 80 warmup and 160 measurement applications. Each performance cell again contains 20 measured requests per class and uses the 32 s denominator. No failed run or request is omitted.

Within these blocks, allowed-minus-excluded short TPOT is 6.730061/7.131971 ms, and long TTFT improves by 16.575631/16.925501 s (31.147731/31.634278%). Allowed progress loses one short success in block 1 and two in block 2, each solely through the completion-TPOT target; their TTFT and completion targets pass. No long request qualifies. Equation 1 gives goodput changes of $-1/32 = -0.03125$ and $-2/32 = -0.0625$ requests/s. These are within-campaign block contrasts. Comparing the extension with the earlier immediate/empty factorial cells is descriptive across separate campaigns, not a paired estimate of the threshold’s effect.

Table 16: All fixed-threshold performance cells in executed order. Successes are each out of 20; TPOT is the short mean and TTFT the long mean. Mixed steps and maximum observed gate waits include the entire initialized phase.

Block	Progress	Short/long	TPOT (ms)	TTFT (s)	Mixed	Max wait (s)
1	Excluded	20/0	40.685	53.216	8	2.078
1	Allowed	19/0	47.415	36.641	44	2.143
2	Allowed	18/0	48.765	36.578	44	2.114
2	Excluded	20/0	41.633	53.504	8	2.048

In each allowed cell, 11/30 prefixes enter through the observed-deadline branch and 19/30 through an empty set; excluded cells have 2/30 and 28/30, respectively. Maximum waiting exceeds the nominal threshold because synchronous native work delays event-loop observation (Table 16). Allowed cells contain 8090/6520 strict whole-step write witnesses across twelve payloads each, including three from the measurement arrival cohort. Excluded cells have zero native-write intersections. Thus the extension admits work before all whole requests drain and retains measured-cohort sender progress; it does not establish complete short-request protection. Aggregate sender CPU is 3750.982/4614.700 ms under allowed progress and 4037.811/4684.830 ms under exclusion; these totals do not identify a unique mediator.

The anonymous `bounded_pacing_artifact` includes all 256 integer client records, the scientific plan, phase-mechanism summaries, source SHA references and the three exact executed control modules. Its independent reconstruction retains the qualification/warmup distinction, observed admission reasons and threshold overshoot. The same portable-launcher and raw-tensor boundaries as the mechanism artifact apply. No threshold search or optimality claim is attached to this conventional probe.

J GENERAL EVENT AND RESOURCE FORMULATION

The following formulation provides context for the concrete admission experiment. It is a standard feasible-execution model, not a new scheduling theorem or an optimizer evaluated in this study.

The observed boundary motivates controlling three coupled decisions: whole versus split placement, admission and release order, and batch/chunk allocation. For a fixed arrival cohort I over window H , let $q_i(\pi)$ indicate whether request i meets every client SLO under policy π . The serving objective is

$$\max_{\pi \in \Pi} \mathbb{E} \left[\frac{1}{H} \sum_{i \in I} q_i(\pi) \right]. \quad (3)$$

The expectation is over execution variability; online actions use only currently published observations. For operation v of request i , let x_v, y_v be dispatch and confirmed completion, d_{vr} its demand on resource r , and C_r capacity. Precedence edges E include prefill, handoff, installation and decode. A feasible execution satisfies

$$\begin{aligned} x_v &\geq a_i, & x_v &\geq y_w & ((w, v) \in E), \\ \sum_v d_{vr} \mathbf{1}\{x_v \leq t < y_v\} &\leq C_r & (\text{all } r, t). \end{aligned} \quad (4)$$

Memory follows the same interval accounting with byte demands and ownership-release events. Compatible KV representation and model state are prerequisites. Resource release is tied to acknowledgements, not a predicted transfer end. Batch composition and concurrent work affect $y_v - x_v$, so independently minimizing each transfer does not solve Equation 3.

K NATIVE PREFIX WORK AND AN ORDINARY CHUNKING CONTROL

A separate development campaign tests whether immediate prefix entry can be retained while limiting the work entering each source-native step. It crosses matched write exclusion/permission with three work rules: the original budget (N), an ordinary per-request cap of 128 query positions (C), and

a PD-prefix-only cap of 128 (P). Every cell keeps the total step budget at 1024, maximum eight running requests, immediate entry, one prefix lease and one staging slot. The two existing input/output trajectories, routes, arrival times, client targets, model, precision and endpoint quotas remain fixed. This tests one prespecified engineering value, not a tuned chunk-size curve.

Actual intervention. The pinned vendor vLLM v1 scheduler computes proposed work separately for running and waiting requests before allocating KV slots. We insert the same callback at these two sites in every arm; N returns the original value. For proposed work $n_i \geq 0$ and request role r_i , the callback returns

$$\hat{u}_i = \min\{n_i, c_m(r_i)\}, \quad c_m(r) = \begin{cases} 128 & m = C \text{ or } (m = P \wedge r = \text{prefix}), \\ +\infty & \text{otherwise.} \end{cases} \quad (5)$$

The original remaining-token budget, model-position limits, allocation failure, preemption and post-schedule accounting still apply. Consequently actual scheduled query work u_i in one iteration satisfies $u_i \leq \hat{u}_i$ and $\sum_i u_i \leq 1024$; under the single-prefix lease, C/P also give $\sum_{i:r_i=\text{prefix}} u_i \leq 128$. These are local budget invariants, not a client-latency or queue-stability theorem. P preserves whole-request rules, while C also chunks whole prefill. Either can change batch composition, iteration count and observer cost.

The ordinary C callback is CPU-executed against the unmodified vendor method with `long_prefill_token_threshold=128`, covering running/waiting requests and complete work. The native experiment keeps that configuration at zero and realizes C at the same sites, so all three arms incur the callback. Twenty-four local tests cover the vendor method, allocation failure, ownership restoration, campaign guards and analysis; eighteen additional plan/constructor/launch checks precede native execution. Source hashes bind the actual method, and every phase records the installed control and actual query work. No scheduler output is changed after allocation. This is an ordinary chunking comparator, not an implementation of the complete Sarathi-Serve scheduler.

Execution and evidence. Six eight-request qualifications precede two reversed six-cell blocks, each cell having 20 warmup and 40 measurement arrivals. The 768 applications comprise 48 qualification, 240 warmup and 480 measurement arrivals. Qualification requires complete replay/KV integrity, owned cleanup, actual cap exposure, and positive whole-step sender witnesses in each allowed arm. Performance direction is not a validity gate. Every cell is retained, with no native retries. The budget and immediate-entry conditions are checked against actual query updates rather than configuration alone.

Figure 3 in the main paper summarizes native exposure; the complete cells follow.

Table 17: All native-budget performance cells in executed order. N is original work; C caps every request at 128; P caps only PD prefixes at 128. Total step budget remains 1024. Short/long counts are each out of 20; TPOT is the short mean and TTFT the long mean. Writes are strict whole-step witnesses from measurement-cohort payloads; mixed steps contain a measurement-cohort prefix. Neither is restricted to an aligned cross-host wall-clock window.

Block	Writes	Rule	Short/long	TPOT (ms)	TTFT (s)	Goodput	Witnesses	Mixed
1	Excluded	N	20/0	41.478	52.860	0.62500	0	0
1	Allowed	N	0/0	57.395	17.394	0.00000	7439	56
1	Excluded	C	20/0	43.212	60.140	0.62500	0	0
1	Allowed	C	0/0	59.380	35.902	0.00000	2602	144
1	Excluded	P	20/0	41.768	59.346	0.62500	0	0
1	Allowed	P	0/0	59.166	35.674	0.00000	2794	144
2	Allowed	P	0/0	59.758	34.935	0.00000	3246	144
2	Excluded	P	20/0	42.968	60.414	0.62500	0	0
2	Allowed	C	0/0	58.907	34.830	0.00000	4057	144
2	Excluded	C	20/0	43.435	60.959	0.62500	0	0
2	Allowed	N	0/0	57.196	17.691	0.00000	7812	54
2	Excluded	N	20/0	41.438	53.261	0.62500	0	0

Both 128-position rules retain measurement-cohort sender witnesses and mixed execution under allowed progress, but neither rescues a short request. Every excluded cell qualifies 20 short requests,

every allowed cell qualifies zero, and every long request fails the joint target (Table 17). Thus the prespecified screen—preserve short successes and reduce long mean TTFT by at least 20% versus matched original exclusion—fails in both blocks. Original allowed progress gives long mean TTFT 17.394/17.691 s; ordinary chunking gives 35.902/34.830 s, and prefix-only chunking gives 35.674/34.935 s. This value of the prefix-only rule does not provide an engineering advance. Small differences between the two capped rules do not establish an advantage over an ordinary control; independent workload confirmation is absent.

Interpretation and reproducibility. A per-step query limit constrains neither the number of mixed steps experienced by a request nor their accumulated host residence. Smaller steps can require additional iterations and change which requests coexist. The per-request analysis intersects native steps only with the same BI clock’s first-visible-to-terminal output window; client TPOT remains on its separate client clock. The resulting exposure statistics are descriptive, and are never subtracted from client latency to manufacture a counterfactual.

The `native_budget_artifact` exports all 768 integer client trajectories, scientific plans, actual per-step work, phase summaries and four exact executed control modules. A separate standard-library reconstruction checks every client outcome with rational TPOT. Native source, phase and client hashes preserve provenance; the export is not a replacement for images, weights, original KV evidence or the private deployment controller. S16 is development evidence on consumed trajectories, not independent workload confirmation. The historical W observations do not become same-campaign baselines for this experiment.

L PROSPECTIVE MICROBURST ARRIVAL BOUNDARY

The matched progress–entry intervention was frozen on one previously unexecuted arrival vector after the evenly spaced and native-budget campaigns. For arrival index i , its offset in seconds is

$$a_i - a_0 = 3.2 \lfloor i/4 \rfloor + 0.01(i \bmod 4). \quad (6)$$

Four alternating short/long requests arrive 10 ms apart every 3.2 s. The first 20 performance arrivals occur in $[0, 16)$ s and the next 40 in $[16, 48)$ s, retaining the 32 s denominator and average measurement arrival rate of 1.25 requests/s. Planned arrivals are integer nanoseconds; actual late offers remain charged from that time. Every phase drains completely. The consumed texts, replay, routes, targets, resources and serial staging remain unchanged. No native-budget cap is installed.

Design and estimand. EI/AI exclude/allow writes with immediate entry; EE/AE exclude/allow writes with current-whole-empty entry. Four eight-request qualifications precede three initialized four-cell blocks, ordered EI–AI–EE–AE, AE–EE–AI–EI, and EE–EI–AE–AI. The first two reverse order; the third is rotated, not a complete Latin square. There are 752 applications: 32 qualification, 240 warmup and 480 measurement. All plans and the analysis were frozen before qualification. Each planned native cell executes once; all native integrity and cleanup checks pass.

The primary quantity is the blockwise interaction in Equation 2. The prespecified directional screen requires a negative interaction and fewer AI than EI short successes in each of all three blocks. It is a finite-block directional check, not an independent-request significance test. All planned cells are retained regardless of performance direction. Empty-set admission tests current work; later arrivals may create mixed steps, so zero mixed execution is not an imposed validity requirement.

Table 18: All microburst performance cells in executed order. Each class has 20 measurement arrivals per cell. TPOT is the short mean and TTFT the long mean; qualification uses every individual request. Witnesses count positive whole-step writes by measurement-cohort payloads. Mixed steps contain a measurement-cohort prefix, irrespective of source wall-clock position.

Block	Cell	Short/long	TPOT (ms)	TTFT (s)	Goodput	Witnesses	Mixed
1	EI	20/0	41.548	53.645	0.62500	0	0
1	AI	0/0	54.289	20.153	0.00000	6710	48
1	EE	20/0	38.958	56.405	0.62500	0	0
1	AE	20/0	38.136	55.726	0.62500	0	0
2	AE	20/0	38.172	56.697	0.62500	0	0
2	EE	20/0	38.085	55.453	0.62500	0	0
2	AI	0/0	54.244	20.039	0.00000	6607	48
2	EI	20/0	41.230	51.829	0.62500	0	0
3	EE	20/0	38.948	56.235	0.62500	0	0
3	EI	20/0	40.398	50.796	0.62500	0	0
3	AE	20/0	39.078	56.063	0.62500	0	0
3	AI	0/0	53.431	19.782	0.00000	8147	48

The three short-TPOT interactions are -13.563 , -12.927 , -12.904 ms. The prespecified directional screen passes in all three blocks. This statement concerns the fixed evaluated cohorts and arrival vector.

Table 19: Blockwise native progress–entry contrasts under microburst arrivals. D_i/D_e are allowed-minus-excluded short-TPOT differences for immediate/empty-set entry. Recovered counts refer to EI successes lost by AI and recovered by AE. Long cost is AE minus AI mean TTFT.

Block	D_i (ms)	D_e (ms)	I (ms)	Recovered	Long cost (s)
1	12.741	-0.822	-13.563	20	35.573
2	13.014	0.087	-12.927	20	36.658
3	13.033	0.130	-12.904	20	36.281

For AE, measurement-cohort socket-write witnesses are 0/0/0, while full-phase mixed-step counts are 0/0/0. Warmup payloads with whole-step write witnesses number 0/1/0. These distinctions determine whether protection retains progress involving the measured payload cohort.

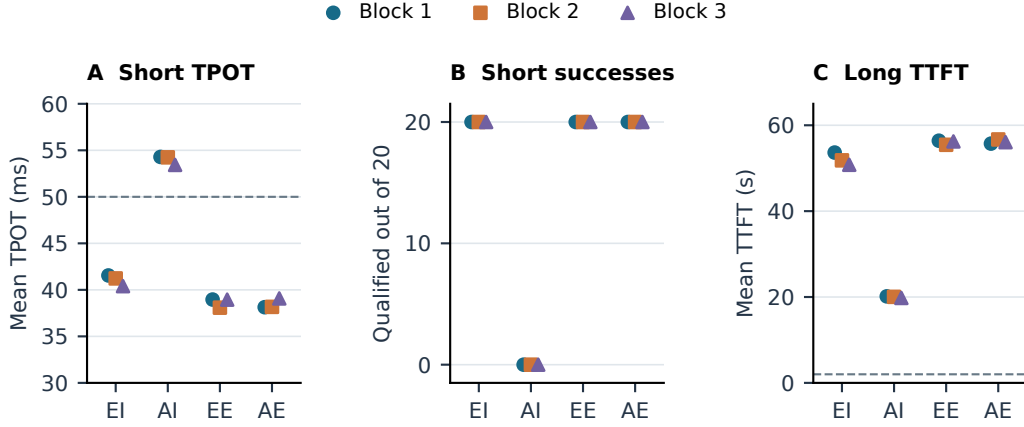


Figure 6: **A new arrival vector tests the progress–entry boundary.** The same four cells are evaluated in three initialized blocks with fixed four-request microbursts. Markers denote block means in A/C and individual-request success counts in B; horizontal dashes in A/C show the fixed latency targets. Mean TPOT alone does not determine qualification. EI/AI use immediate entry and EE/AE current-whole-empty entry; E/A exclude/allow native writes. The dots are measured cells, not confidence intervals or a continuous performance frontier.

A posthoc alignment on the single source clock places the first measurement-prefix admission under AE +14.636/+15.143/+14.727 s relative to the final measured-short terminal event. With AI, 12/12/12 of the twenty measurement prefixes enter before that terminal event. Positive offsets mean admission after short completion; this describes execution ordering rather than a new prespecified efficacy test.

Across the twelve performance phases, maximum offer/dispatch lateness is 4.930/5.052 ms. After the 48 s arrival-window endpoint, 19–32 of each phase’s 60 applications remain to complete; full drain lasts 26.507–61.507 s. All late completion remains included in the fixed measurement cohort, rather than changing the goodput denominator.

Scope and reconstruction. This tests one new arrival condition, not new texts, a new length distribution, natural-answer quality or multiple staging slots. Sender witnesses are assigned by the payload’s arrival cohort. A warmup payload can coexist with measured work; source-native and client clocks are never subtracted. Host-step residence includes runtime and synchronization work and is not exclusive GPU time. The three blocks share workloads and hardware, and requests within them interact. No population confidence interval or sustained-capacity estimate is inferred.

The anonymous `arrival_boundary_artifact` retains all 752 integer client records, exact arrival offsets, routes, replay and executed controls, relative source steps, admission receipts and all measured-short output windows. Its standard-library `summarize.py` verifies the burst vector, chronology, full completion and frozen targets and reconstructs client qualification using rational TPOT. `recompute_arrival_boundary.py` independently reconstructs native mixed-step exposure and separates warmup/measurement witnesses. Images, weights, raw KV tensors and host ownership records are separate. W was not rerun in this campaign, so this evidence does not rank a new service algorithm against W.

M REPLAY PUBLICATION AT THE NATIVE CPU BOUNDARY

Replay implementation can affect the observed runtime. We test one specific alternative explanation: whether an extra sampler-to-host-to-device roundtrip is necessary for the progress-associated short-request penalty. This is a placement intervention within complete-output replay; natural-generation quality is not its target.

Locate the existing synchronization. The pinned source runner first samples native token IDs. It later materializes those IDs as CPU lists before discarding partial-prefill samples and updating input-batch tokens and request output histories. Legacy replay extracts raw choices in the sampler wrapper, replaces them with a device tensor of reference tokens, and then reaches that native CPU conversion. The alternative leaves native sampling intact and rewrites the already-materialized CPU list, before the same discard and history updates. Whole requests retain their exact reference trajectory; prefix requests retain the natural extra sample. Input IDs, positions, raw choices, native EOS, complete work, KV integrity, transport ownership and installation checks remain enabled.

Only single-rank, non-speculative, one-token sampling is supported. A hash-bound, single-assignment source transformation installs an instance-local hook; it does not modify the vendor file. Closing the worker restores the original dispatch. Seventeen CPU tests cover the pinned method transformation, decorator, lane reordering, partial-sample discard, replay history, EOS, failure before partial publication and restoration. Twelve actual plan/constructor/launch checks precede native qualification. Every native query lane is subsequently checked against its actual input and committed history.

Factorial design. EL/AL exclude/allow writes with legacy sampler replay; EC/AC exclude/allow writes with CPU-list replay. All cells use immediate prefix entry, the common owned sender and native mutex, and the original regular-arrival workload, fixed destinations, FP16 model, resources and targets. Four eight-request qualifications precede two separately initialized blocks in orders EL–AL–EC–AC and AC–EC–AL–EL. Both allowed qualifications must exhibit a positive whole-step write witness; excluded writes must not overlap native execution. All 512 applications are retained: 32 qualification, 160 warmup and 320 measurement. No native cell is repeated or selected by outcome.

With $Y_{p,r}^{(b)}$ denoting mean short completion TPOT, progress $p \in \{E, A\}$ and replay placement $r \in \{L, C\}$, the prespecified contrasts are

$$D_r^{(b)} = Y_{A,r}^{(b)} - Y_{E,r}^{(b)}, \quad J^{(b)} = D_C^{(b)} - D_L^{(b)}. \quad (7)$$

The descriptive sensitivity screen requires $D_L^{(b)} > 0$ and $|D_C^{(b)}| \leq |D_L^{(b)}|/2$ in both blocks. This tests a substantial attenuation, not statistical equivalence or a population confidence bound. All cells report individual joint qualification, long TTFT and goodput as well as short TPOT.

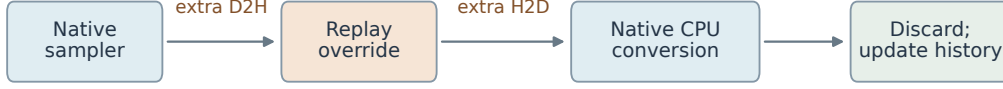
Table 20: All replay-placement performance cells in executed order. Each class has 20 measured requests per cell. Short TPOT and long TTFT are class means; qualification uses the joint target on individual requests.

Block	Cell	Short/long qualified	Short TPOT (ms)	Long TTFT (s)	Goodput
1	EL	20/0	41.629	53.517	0.62500
1	AL	0/0	57.482	16.977	0.00000
1	EC	20/0	41.361	52.714	0.62500
1	AC	0/0	57.838	17.122	0.00000
2	AC	0/0	56.788	16.524	0.00000
2	EC	20/0	41.267	53.352	0.62500
2	AL	0/0	58.135	17.025	0.00000
2	EL	20/0	41.564	52.949	0.62500

Legacy progress contrasts are 15.853/16.571 ms; CPU-list contrasts are 16.477/15.521 ms. The placement interactions are +0.623/-1.050 ms. The prespecified attenuation screen does not pass across both blocks. All native integrity and owned cleanup checks pass.

A Move replay to the existing CPU boundary

Legacy



CPU list



B Matched sender-progress contrasts

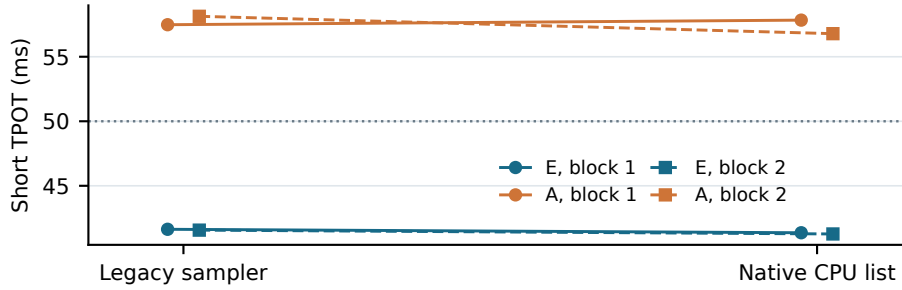


Figure 7: **Test the replay boundary that the runtime actually uses.** (A) Legacy replay adds a raw-ID D2H and forced-ID H2D before the native CPU conversion. CPU-list replay writes at that existing conversion boundary. Both update histories after partial-prefill discard and retain required checks. (B) Dots are separately initialized cell means; line styles distinguish blocks. The dashed line is the short TPOT target. These two blocks do not provide a population confidence interval.

Interpretation and limits. The intervention removes the additional early raw-ID readback and forced-token device roundtrip while changing the CPU replay loop’s placement. It does not isolate those costs from one another. Both arms retain native CPU materialization and common validation; neither is an uninstrumented runtime. Source-step residence also includes host runtime work. A surviving progress penalty therefore excludes this extra roundtrip as a necessary condition in the tested configuration, without identifying a unique GPU or CPU mediator.

The anonymous `replay_boundary_artifact` preserves all integer client events, actual controls, relative source steps, measured-short output windows, source provenance and the reconstruction program. Export independently checks exact rational client TPOT and every native query/history lane before publication. The two initialized blocks use consumed trajectories and serial staging. W is not rerun, and this implementation-sensitivity experiment does not establish a new serving algorithm or SOTA.